# Food Grain Distribution in Gujarat: Summary

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**The decision**

Allocate wheat and rice from four depots to six districts at minimum cost, within stock and loading limits.

**Model and method**

- 48 shipment variables; 24 constraints.
- Two-Phase Simplex: 56 pivots; same cost as SciPy HiGHS.

**Why Surat needs four depots**

| Supply depot | Surat allocation (MT) |
|---|---|
| Sabarmati | 11,000 |
| Chhani | 7,000 |
| Ghanteshwar | 12,000 |
| Gandhidham | 4,000 |

**Main results**

| | |
|---|---|
| Demand met | **132,000 MT** |
| Monthly cost | **Rs. 5.6983 crore** |
| Average cost | **Rs. 431.69/MT** |
| Active routes | **9 of 24** |

**Chhani loading-capacity expansion**

- Raise monthly capacity from 28,000 to 34,000 MT.
- Save **Rs. 55.05 lakh/month**, or **Rs. 6.61 crore/year**.
- Baseline shadow price: Rs. 1,022.90/MT; no benefit beyond 34,000 MT stock.

**Recommendation:** Compare loading-capacity expansion costs with the estimated savings.

Inputs: assumed stocks, demand, loading capacity and freight rates; OSRM road distances.